Filipe Oliveira Barroso received his Ph.D. degree in Biomedical Engineering from University of Minho (Portugal) in 2016. He is currently **Ramón y Cajal Postdoctoral Fellow** (he **ranked 2nd** in the area of knowledge 'Industrial Production, Civil Engineering and Engineering for Society', being awarded with the **Excellence Grant RyC-MaX** from CSIC). Dr. Oliveira Barroso was also awarded the **R3 Certificate** as an established researcher in 2023, followed by the Grant Award **Consolidación Investigadora** 2024. In 2023, he was awarded the prestigious **Postdoctoral Junior Leader Retaining fellowship from "la Caixa" Foundation** (he **ranked 1st** in the area of knowledge 'Biomedical Engineering and Technology'). In 2022, he was awarded with the highly competitive Juan de la Cierva – Incorporación Postdoctoral fellowship.

Since December 2023, he has been the **Group Leader** of the **Neural Engineering Lab** (Cajal Neuroscience Center - CSIC). The main goal of his research is to design breakthrough neural engineering interventions aiming at promoting neurorehabilitation. To achieve this goal, he uses neural interfaces and techniques able to modulate and strengthen neural pathways (*e.g.*, he coordinated an international study that achieved prolonged (up to 24h) reduction of pathological tremor using customized peripheral electrical stimulation). Dr. Oliveira Barroso was the **Coordinator and Principal Investigator of the EU Project EXTEND** "Bidirectional Hyper-Connected Neural System" (Grant Agreement Number 779982) from 2019 to 2022.

From 2016 to February 2018, Dr. Barroso was a Postdoctoral Researcher at **Feinberg School of Medicine - Northwestern University (Chicago, IL, U.S.A.)**, where he enriched his knowledge on neural engineering, neuroplasticity, motor control, brain-machine interfaces and spinal cord injury, namely by conducting research for the Projects "The Neural Control of Internal Joint State" (NIH Project number 5R01NS086973-04) and "Development of an FES neuroprosthesis for rehabilitation following SCI" (Senior Research Grant 340943 from the Craig H. Neilsen Foundation) under the supervision of Professors Matthew C. Tresch and Lee E. Miller, respectively.

His interests include neural engineering, neurorehabilitation, tremor, AI, movement disorders, motor control, electromyography, wearable robots, biomechanics and muscle synergies. Within these areas, he has (co)-authored **33 papers** in peer-reviewed Journals indexed in Journal Citation Reports (JCR) of ISI Web of Knowledge (**20 articles in Journals of the first quartile, 7 as first or co-first author, 3 as senior author, 7 as corresponding author**) and over 65 among conference papers/abstracts, books and book chapters, which have currently received cumulatively **1444 citations** (according to Google Scholar) and an **h-index of 20**. He has participated in **14 R&D projects** (2 Europeans, 2 from U.S.A., 1 from Japan and 9 Spanish national projects). He has also **supervised 6 Doctoral Theses** (4 already defended) 1 Master's Thesis and 13 Bachelor Theses (11 already defended). Dr. Barroso was also the supervisor of 2 JAE Intro grants offered by CSIC, aimed at university students interested in starting a research career.

As part of his service to the scientific community, he has been referee for several Journals (including high-impact journals such as Nature Communications, Journal of NeuroEngineering and Rehabilitation, Journal of Neural Engineering, among others) and International Conferences. Dr. Oliveira Barroso is Associate Editor for 1 scientific journal, Review Editor for 3 scientific journals and he was also Guest Editor of the Special Issue "Neurorobotics and Neuroprostheses: Towards a new anatomy" in the scientific journal The Anatomical Record. He has **1 USA patent** (No. 17/926,278, Title "Control method for a neuroprosthetic device for the reduction of pathological tremors") based on International Application PCT/EP2021/063189, which was already licensed, and **1 European patent** (Title "System for monitoring and reducing pathological tremor in a user"). Dr. Barroso is also IEEE Senior Member.